LESSON 3.8

Solve Problems with Unit Rates

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by using unit rates to compare options.

WHY THIS MATTERS You're the new supply manager for the arcade! You need to order tokens, prizes, and snacks from different suppliers. Each supplier offers a different package deal. Your job is to find the unit rate for each deal and figure out which supplier gives you the best value for every item you need to order. Fun Supply Co. sells 500 tokens for $60. Arcade Depot sells 350 tokens for $38.50. Which supplier gives the better value for tokens?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Unit Rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	$2.50 per pound means $2.50 for every 1 pound
Better Buy *Spanish: Mejor compra*	The choice that costs less for each item.	$0.10 each vs $0.15 each — the $0.10 option is the better buy
Comparison *Spanish: Comparación*	Looking at two or more amounts to see how they are related.	$3 per game < $4 per game
Per Unit *Spanish: Por unidad*	For each single item.	$0.50 per token means each token costs $0.50
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	120 miles in 3 hours is a rate (miles and hours are different units)
Proportion *Spanish: Proporción*	A math sentence saying two ratios are equal. It helps find a missing number.	5/3 = x/12 → x = 20

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Fun Supply Co. sells 500 tokens for $60. I want the cost for 1 token.

1. I divide the price by the tokens: $60 ÷ 500 = $0.12.
2. So this deal is $0.12 per token. That is the unit rate.
3. Now I can compare it fairly to a deal with a different number of tokens.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Arcade Depot sells 350 tokens for $38.50. What do we divide?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I find the better buy by comparing the _____ for each option. The option with the _____ unit rate is the better buy because _____.

Word bank Unit Rate | Better Buy | Comparison | Per Unit | Rate | Proportion | unit rate | lower | divide | cost | per | compare

WATCH OUT!

A common mistake

A common mistake in Solve Problems with Unit Rates is comparing the total prices instead of the cost per item

LESSON 3.8

Solve Problems with Unit Rates

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the unit rate for each deal to determine which is cheaper.

Item	Deal	Unit Rate
Erasers Pack A	12 for $3.60	$0.30 each
Erasers Pack B	8 for $2.80	
Erasers Pack C	20 for $5.00	

HINT Pack A's finished row shows the move: 12 for $3.60 became $3.60 ÷ 12 = $0.30 each.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which is the better buy: 6 bouncy balls for $4.50 or 10 bouncy balls for $8.00?

(A) 10 for $8.00 — $0.80 each

(B) They cost the same per ball

(C) Not enough information

(D) 6 for $4.50 — $0.75 each

HINT You can't compare $4.50 and $8.00 straight across — the packs hold different numbers of balls.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A faucet drips 15 ounces of water in 5 minutes. What is the unit rate?

(A) 0.33 ounces per minute

(B) 3 ounces per minute

(C) 5 ounces per minute

(D) 15 ounces per minute

HINT The unit rate here is ounces per ONE minute — 15 ounces spread across 5 minutes.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

A printer prints 42 pages in 6 minutes. How many pages does it print in 15 minutes at the same rate?

(A) 70 pages

(B) 90 pages

(C) 105 pages

(D) 120 pages

HINT This takes two steps: first find pages per ONE minute, then scale up to 15 minutes.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Store A sells 5 notebooks for $8. Store B sells 3 notebooks for $5.25. Which has the lower price per notebook?

(A) Store B ($1.75 each)
(B) Store A ($1.60 each)
(C) Same price
(D) Cannot determine

HINT Each store sells a different number of notebooks, so compare the price of ONE notebook at each store.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** Which gum deal is the better buy: 4 packs for $2.00 or 10 packs for $4.50?
2. **Attempt** A student says: "$2.00 is less than $4.50, so 4 packs for $2.00 is the better buy."
3. **Check** The student compared the total prices instead of the cost for one pack.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT The packages hold different numbers of packs, so a smaller total price does not automatically win.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

SENTENCE STARTER I find the better buy by comparing the ___ for each option. The option with the ___ unit rate is the better buy because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

LESSON 3.8

Solve Problems with Unit Rates

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the unit rate for each deal and identify the better buy.

Option	Deal	Unit Rate
Store A Batteries	8 for $6.00	
Store B Batteries	12 for $8.40	
Store A Candy	15 for $6.75	
Store B Candy	20 for $8.00	

SHOW YOUR WORK

_____

_____

2 **Write the name of the correct group on each line.**

Three friends each walked different distances in different times. Sort them from fastest to slowest.

Groups: Fastest | Middle | Slowest

_____ Aiden: 6 miles in 2 hours

_____ Bella: 10 miles in 4 hours

_____ Carlos: 4 miles in 2 hours

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A machine dispenses 24 tokens in 3 minutes. At this rate, how many tokens does it dispense in 10 minutes?

(A) 60 tokens
(B) 72 tokens
(C) 80 tokens
(D) 240 tokens

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A factory produces 360 widgets in 8 hours. At the same rate, how many widgets are produced in a 12-hour shift?

(A) 360 widgets
(B) 480 widgets
(C) 540 widgets
(D) 720 widgets

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** A vending machine dispenses 24 tokens in 3 minutes. How many tokens per minute is that?
2. **Attempt** A student writes: 3 ÷ 24 = 0.125, so the machine dispenses 0.125 tokens per minute.
3. **Check** The student divided minutes by tokens, which gives minutes per token, not tokens per minute.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.8

Solve Problems with Unit Rates

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Two delivery drones fly at different rates. Drone A travels 9 miles in 12 minutes. Drone B travels 7 miles in 10 minutes. Complete the table and determine which drone delivers faster.

Drone	Distance	Time	Miles per Minute	Faster?
Drone A	9 miles	12 min		
Drone B	7 miles	10 min		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Unit Rate Error

1. **Which is the better buy: 5 lb for $8.75 or 3 lb for $5.10?** Compare two deals
2. **Find unit rate for Deal 1** 5 ÷ $8.75 = 0.57 lb per dollar
3. **Find unit rate for Deal 2** 3 ÷ $5.10 = 0.59 lb per dollar
4. **Compare** 0.59 > 0.57, so Deal 2 is the better buy

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Two arcade supply companies offer token deals. Company X: 300 tokens for $36. Company Y: 500 tokens for $55. The arcade needs 1,000 tokens. Find the unit rate for each company, determine which is cheaper per token, and calculate the total cost of buying 1,000 tokens from the better deal.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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